Solving linear systems of DE

Ex: $\begin{cases} x' - 2y = 4t & (1) \\ y' + 2y - 4x = -4t - 2 & (2) \end{cases}$ $\begin{matrix} x(0) = 4 \\ y(0) = -5 \end{matrix}$ $\begin{matrix} x(t) - ? \\ y(t) - ? \end{matrix}$

I. (from algebra)

from (1) find $y = \dfrac{x' - 4t}{2}$ (3) ← sometimes not so easy

sub in (2) $\Rightarrow \dfrac{x'' - \cancel{4}}{2} + x' - \cancel{4t} - 4x = -\cancel{4t} - \cancel{2}$

$\underline{x'' + 2x' - 8x = 0}$ ← 2^{nd} order lin. hom.

$x(0) = 4$ $x'(0) \overset{(1)}{=} 4 \cdot 0 + 2 \cdot y(0) = 2 \cdot y(0) = -10$

find $x(t)$, from (3) find $y(t)$.

II. $$\begin{cases} x' - 2y = 4t \\ y' + 2y - 4x = -4t - 2 \end{cases} \qquad \begin{aligned} x(0) &= 4 \\ y(0) &= -5 \end{aligned}$$

II. (Laplace transform).

$$\begin{cases} \mathcal{L}\{x'\} - 2Y = 4\mathcal{L}\{t\} \\ \mathcal{L}\{y'\} + 2Y - 4X = -4\mathcal{L}\{t\} - 2\mathcal{L}\{1\} \end{cases}$$

$$\begin{cases} sX - 4 - 2Y = 4 \cdot \frac{1}{s^2} \\ \underline{sY} + 5 + \underline{2Y} - \underline{\underline{4X}} = -4\frac{1}{s^2} - 2\frac{1}{s} \end{cases}$$

$$\Rightarrow \begin{cases} sX - 2Y = \frac{4}{s^2} + 4 \\ -4X + (s+2)Y = -\frac{4}{s^2} - \frac{2}{s} - 5 \end{cases}$$

$$\begin{cases} sX - 2Y = \frac{4}{s^2} + 4 \\ -4X + (s+2)Y = -\frac{4}{s^2} - \frac{2}{s} - 5 \end{cases}$$

$$\boxed{y = \frac{x' - 4t}{2}}$$

$$\Rightarrow \begin{cases} s(s+2)X - 2(s+2)Y = \frac{(4+4s^2)(s+2)}{s^2} \\ -8X + 2(s+2)Y = \frac{2(-4-2s-5s^2)}{s^2} \end{cases} + \Rightarrow$$

$$\Rightarrow (s(s+2)-8)X = \frac{(4+4s^2)(s+2) + 2(-4-2s-5s^2)}{s^2}$$

$$\Rightarrow X(s) = \frac{4s-2}{(s+4)(s-2)} = \frac{3}{s+4} + \frac{1}{s-2}$$

$$\Rightarrow x(t) = \mathcal{L}^{-1}\{X\}(t) = 3e^{-4t} + e^{2t}$$

$$\Rightarrow y = \frac{x' - 4t}{2} = \frac{-12e^{-4t} + 2e^{2t} - 4t}{2} = -6e^{-4t} + e^{2t} - 2t$$

Interconnected fluid tanks (Sec. 5.1)

Two large tanks, each holding 24 liters of a brine solution, are interconnected by pipes as shown in Figure 5.1. Fresh water flows into tank A at a rate of 6 L/min, and fluid is drained out of tank B at the same rate; also 8 L/min of fluid are pumped from tank A to tank B, and 2 L/min from tank B to tank A. The liquids inside each tank are kept well stirred so that each mixture is homogeneous. If, initially, the brine solution in tank A contains x_0 kg of salt and that in tank B initially contains y_0 kg of salt, determine the mass of salt in each tank at time $t > 0$.[†]

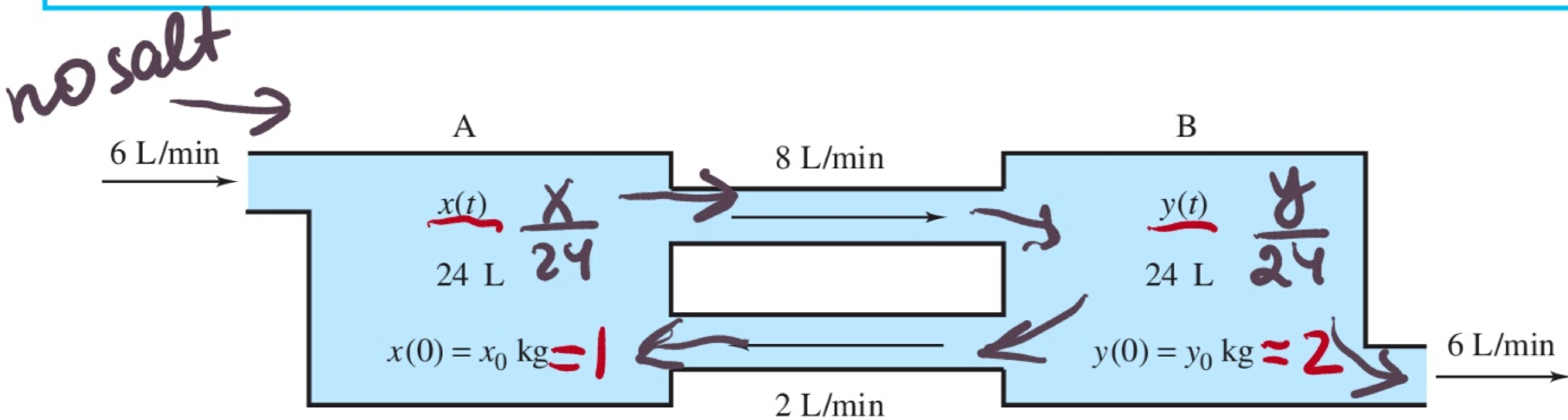

Figure 5.1 Interconnected fluid tanks

$$\frac{dx}{dt} = \text{input} - \text{output}$$

$$A: \frac{dx}{dt} = 0 + \frac{y\,\text{kg}}{24\,\cancel{L}} \cdot 2\,\frac{\cancel{L}}{\text{min}} - \frac{x}{24} \cdot 8$$

$$B: \frac{dy}{dt} = \frac{x}{24} \cdot 8 - \frac{y}{24} \cdot 8$$

$$\Rightarrow \begin{cases} x' = \dfrac{y}{12} - \dfrac{x}{3} & x(0) = 1 \\ y' = \dfrac{x}{3} - \dfrac{y}{3}, & y(0) = 2 \end{cases}$$

$$\begin{cases} x' = -\frac{1}{3}x + \frac{1}{12}y \\ y' = \frac{1}{3}x - \frac{1}{3}y \end{cases}, \quad x(0)=1, \quad y(0)=2$$

$$\begin{cases} sX - 1 = -\frac{1}{3}X + \frac{1}{12}Y \\ sY - 2 = \frac{1}{3}X - \frac{1}{3}Y \end{cases} \Rightarrow \begin{cases} \left(s+\frac{1}{3}\right)X - \frac{1}{12}Y = 1 \\ -\frac{1}{3}X + \left(s+\frac{1}{3}\right)Y = 2 \end{cases} \Rightarrow$$

$$\Rightarrow X = 3\left(\left(s+\frac{1}{3}\right)Y - 2\right) = (3s+1)Y - 6$$

$$\left(s+\frac{1}{3}\right)(3s+1)Y - 6\left(s+\frac{1}{3}\right) - \frac{1}{12}Y = 1$$

$$\Rightarrow \left(3\left(s+\frac{1}{3}\right)^2 - \frac{1}{12}\right)Y = 1 + 6s + 2 = 6s+3 \Rightarrow$$

$$\Rightarrow \cancel{3}\left(\left(s+\frac{1}{3}\right)^2 - \frac{1}{36}\right)Y = \cancel{3}(2s+1) \Rightarrow \left(s+\frac{1}{3}-\frac{1}{6}\right)\left(s+\frac{1}{3}+\frac{1}{6}\right)Y = 2s+1$$

$$\left(s+\frac{1}{6}\right)\left(s+\frac{1}{2}\right)Y = 2s+1 \Rightarrow Y = (2s+1)\Big/\left(s+\frac{1}{6}\right)\left(s+\frac{1}{2}\right)$$

$$Y = \frac{2s+1}{\left(s+\frac{1}{6}\right)\left(s+\frac{1}{2}\right)} = \frac{2\cancel{\left(s+\frac{1}{2}\right)}}{\left(s+\frac{1}{6}\right)\cancel{\left(s+\frac{1}{2}\right)}} = \frac{2}{s+\frac{1}{6}}$$

$$\Rightarrow \quad y(t) = \mathcal{L}^{-1}\{Y\}(t) = \mathcal{L}^{-1}\left\{\frac{2}{s+\frac{1}{6}}\right\}(t) = \underline{2e^{-\frac{1}{6}t}}$$

$$y' = \frac{1}{3}x - \frac{1}{3}y \quad \Rightarrow \quad x = 3\left(y' + \frac{1}{3}y\right) = 3y' + y$$

$$x = 3 \cdot 2\left(-\frac{1}{6}\right)e^{-\frac{1}{6}t} + 2e^{-\frac{1}{6}t} = \underline{e^{-\frac{1}{6}t}}$$

Electrical systems (sec. 5.7)

At time $t = 0$, the charge on the capacitor in the electrical network shown in Figure 5.38 is 2 coulombs (C), while the current through the capacitor is zero. Determine the charge on the capacitor and the currents in the various branches of the network at any time $t > 0$.

Kirchhoff's law

$$\boxed{L\frac{dI}{dt} + RI + \frac{1}{C}q = E(t)}$$

$$\boxed{I = \frac{dq}{dt}}$$

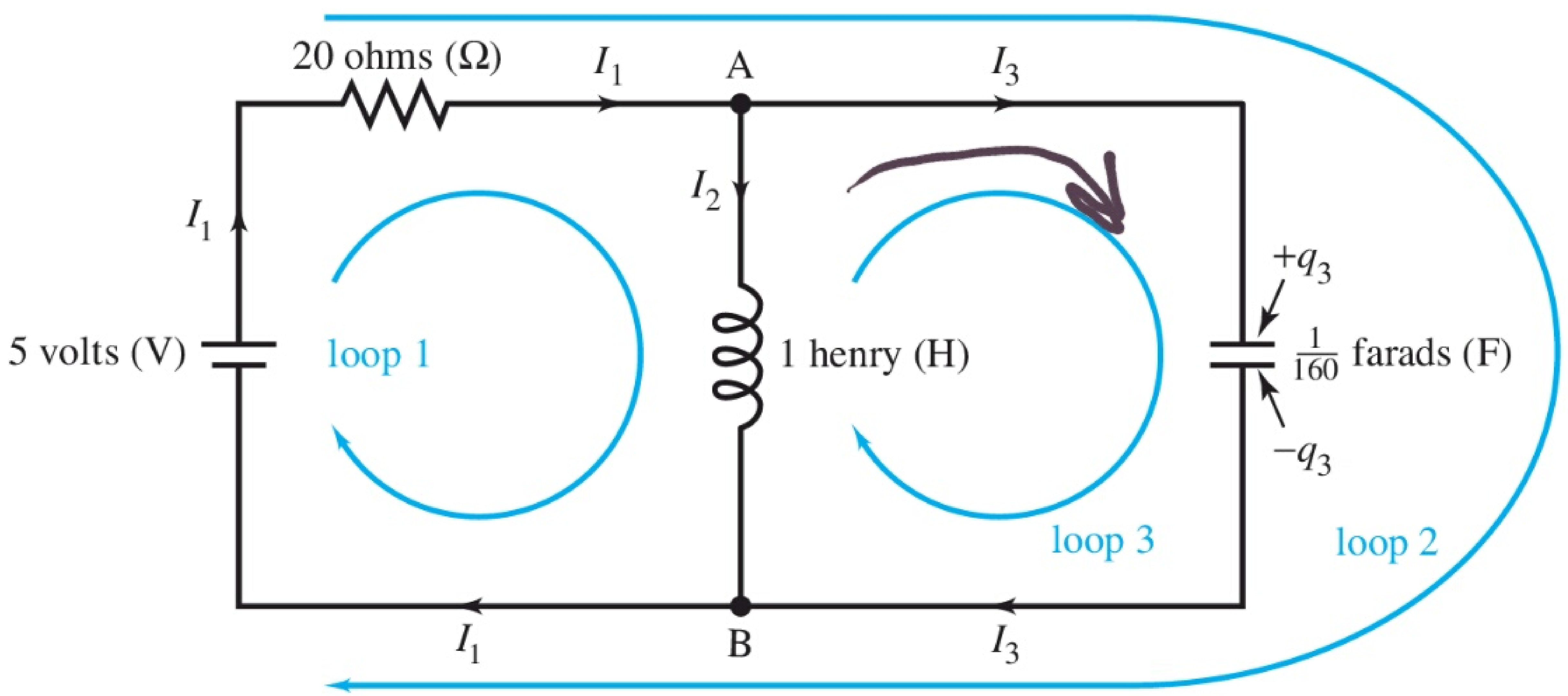

Figure 5.38 Schematic of an electrical network

$$q_3(0) = 2$$

$$I_3(0) = 0$$

$$\text{"} \; q_3'(0)$$

loop1: $1 \cdot \frac{dI_2}{dt} + 20I_1 = 5$

loop2: $20I_1 + 160q_3 = 5$

loop3: $160q_3 - \frac{dI_2}{dt} = 0$

$$I_1 = I_2 + I_3, \quad I_3 = q_3'$$

$$\begin{cases} I_2' + 20 I_1 = 5 \\ 20 I_1 + 160 q_3 = 5 \\ I_1 = I_2 + I_3 \\ I_3 = q_3' \end{cases} \Rightarrow I_1 = I_2 + q_3'$$

$$\boxed{\begin{array}{l} q_3(0) = 2 \\ q_3'(0) = 0 \end{array}}$$

$$I_3 = q_3' = -\frac{80}{3} e^{-4t} \sin 12t$$

$$I_2 = I_1 - I_3 = \frac{1}{4} - 16 e^{-4t} \cos 12t + \frac{64}{3} e^{-4t} \sin 12t$$

$$\begin{cases} I_2' + 20 I_2 + 20 q_3' = 5 \\ 20 I_2 + 20 q_3' + 160 q_3 = 5 \end{cases} \Rightarrow I_2 = \frac{1}{20}\left(5 - 20 q_3' - 160 q_3\right) = \frac{1}{4} - q_3' - 8 q_3$$

$$-q_3'' - 8 q_3' + \cancel{5} - \cancel{20 q_3'} - 160 q_3 + \cancel{20 q_3'} = \cancel{5}$$

$$q_3'' + 8 q_3' + 160 q_3 = 0 \overset{\text{find}}{\Longrightarrow} q_3(t) = 2 e^{-4t} \cos 12t + \frac{2}{3} e^{-4t} \sin 12t$$

find $I_1 = \dfrac{5 - 160 q_3}{20} = \dfrac{1}{4} - 16 e^{-4t} \cos 12t - \dfrac{16}{3} e^{-4t} \sin 12t$

Laplace transform summary

1. Definition of LT
2. $\mathcal{L}\{af+bg\} = a\mathcal{L}\{f\} + b\mathcal{L}\{g\}$.

all other properties will be given to you in the table similar to the one at the last page of the book

you must learn how to use this table!